

If a primary alkyl halide and a chiral tertiary halide both undergo substitution reactions, the reaction with the tertiary halide will have more products because:

- ☐ A. the primary halide undergoes an S_N2 reaction, which occurs in one step and is not stereospecific. [10%]
- ✓ ☒ B. the chiral tertiary halide undergoes an S_N1 reaction, which forms a carbocation and is not stereospecific. [79%]
- ☐ C. the nucleophile can add to the electrophile from either side of the primary alkyl halide. [5%]
- ☐ D. the backside of the electrophilic carbon on a tertiary halide is not sterically hindered. [4%]

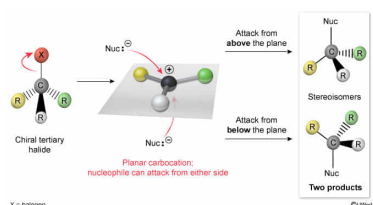
Correct

78% answered correctly

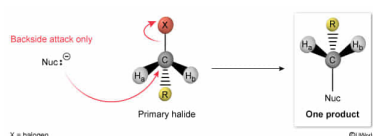
Explanation:

Substitution reactions occur in one of two ways: an S_N1 or an S_N2 mechanism. These substitution reactions differ in number of steps and stereospecificity: An S_N1 reaction occurs in two steps and is *not* stereospecific whereas an S_N2 reaction occurs in a single step and *is* stereospecific. **Stereospecific reactions** occur when **stereoisomers** that undergo the same reaction each give only one specific stereoisomer of the product. The stereochemistry of the starting material determines the stereochemistry of the product.

The first step of an S_N1 reaction involves breaking the bond between the leaving group and the adjoining carbon, forming a **planar carbocation** intermediate. In the second step, the nucleophile can add to either the top or bottom of the planar carbocation, leading to the formation of **two products**.



In contrast, S_N2 reactions occur in a single step and the nucleophile always attacks on the side opposite the leaving group (backside attack), causing an **inversion** at the electrophilic carbon. An S_N2 reaction **forms only one product**.



Tertiary alkyl halides form more **stable carbocations** than primary halides because the additional alkyl groups stabilize the carbocation by the inductive effect. Therefore, tertiary alkyl halides readily undergo S_N1 reactions. In contrast, carbocations are less stable on primary alkyl halides; however, the relative lack of steric hindrance allows primary alkyl halides to more readily undergo S_N2 reactions. **Chiral tertiary** alkyl halides form two products (**stereoisomers**) in S_N1 reactions because S_N1 reactions are **not stereospecific**.

(Choices A and C) S_N2 reactions occur in one step, and primary alkyl halides undergo this reaction more readily than the more sterically hindered tertiary halides. However, S_N2 reactions *are* stereospecific, and only one stereoisomer is formed.

(Choice D) The backside of the electrophilic carbon on a tertiary halide is **sterically hindered** by the bulk of the other substituents.

Educational objective:

Stereospecific reactions occur when stereoisomers undergo the same reaction, and each gives only one stereoisomer product. Tertiary alkyl halides readily undergo S_N1 reactions, which are not stereospecific because the nucleophile can attack the planar carbocation intermediate from the top or the bottom of the carbocation, forming one of two stereoisomers. Primary alkyl halides readily undergo S_N2 reactions, which are stereospecific because the nucleophile attacks only from the opposite side, where the leaving group is eliminated (backside attack).

Time spent: 0 sec
Subject:
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